

Try Mermaid Advanced Editor — OSS users get 10% off with code JS26

Get started

Sankey diagram (v10.3.0+)

A sankey diagram is a visualization used to depict a flow from one set of values to another.

WARNING

This is an experimental diagram. Its syntax are very close to plain CSV, but it is to be extended in the nearest future.

The things being connected are called nodes and the connections are called links.

Example

This example taken from [observable](#). It may be rendered a little bit differently, though, in terms of size and colors.

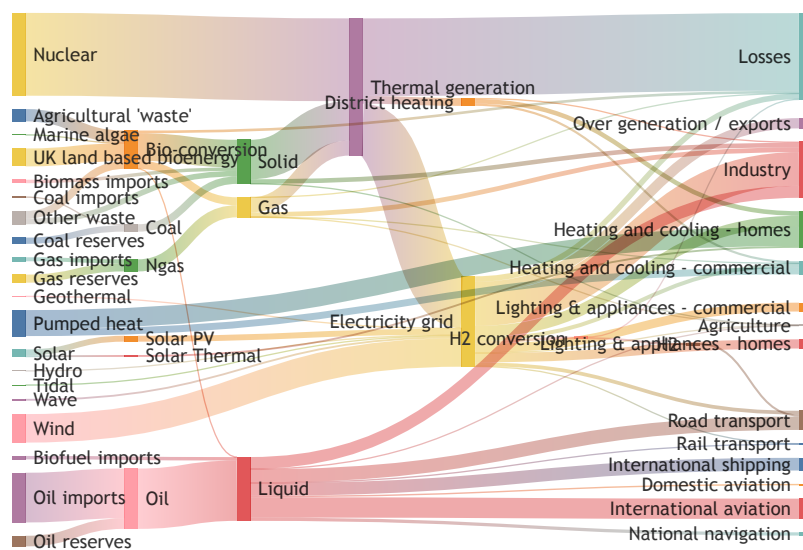
Code:

```
mermaid
graph LR
    subgraph config
        direction TB
        A[config] --> B[sankey]
        B --> C[showValues: false]
    end
    subgraph sankey
        direction TB
        D["Agricultural 'waste',Bio-conversion,124.729"]
        E["Bio-conversion,Liquid,0.597"]
        F["Bio-conversion,Losses,26.862"]
        G["Bio-conversion,Solid,280.322"]
        H["Bio-conversion,Gas,81.144"]
        I["Biofuel imports,Liquid,35"]
        J["Biomass imports,Solid,35"]
        K["Coal imports,Coal,11.606"]
        L["Coal reserves,Coal,63.965"]
        M["Coal,Solid,75.571"]
        N["District heating,Industry,10.639"]
        O["District heating,Heating and cooling - commercial,22.505"]
        P["District heating,Heating and cooling - homes,46.184"]
        Q["Electricity grid,Over generation / exports,104.453"]
        R["Electricity grid,Heating and cooling - homes,113.726"]
        S["Electricity grid,H2 conversion,27.14"]
        T["Electricity grid,Industry,342.165"]
        U["Electricity grid,Road transport,37.797"]
        V["Electricity grid,Agriculture,4.412"]
        W["Electricity grid,Heating and cooling - commercial,40.858"]
        X["Electricity grid,Losses,56.691"]
        Y["Electricity grid,Rail transport,7.863"]
        Z["Electricity grid,Lighting & appliances - commercial,90.008"]
        AA["Electricity grid,Lighting & appliances - homes,93.494"]
        AB["Gas imports,Ngas,40.719"]
        AC["Gas reserves,Ngas,82.233"]
        AD["Gas,Heating and cooling - commercial,0.129"]
    end
```

```

Gas, Losses, 1.401
Gas, Thermal generation, 151.891
Gas, Agriculture, 2.096
Gas, Industry, 48.58
Geothermal, Electricity grid, 7.013
H2 conversion, H2, 20.897
H2 conversion, Losses, 6.242
H2, Road transport, 20.897
Hydro, Electricity grid, 6.995
Liquid, Industry, 121.066
Liquid, International shipping, 128.69
Liquid, Road transport, 135.835
Liquid, Domestic aviation, 14.458
Liquid, International aviation, 206.267
Liquid, Agriculture, 3.64
Liquid, National navigation, 33.218
Liquid, Rail transport, 4.413
Marine algae, Bio-conversion, 4.375
Ngas, Gas, 122.952
Nuclear, Thermal generation, 839.978
Oil imports, Oil, 504.287
Oil reserves, Oil, 107.703
Oil, Liquid, 611.99
Other waste, Solid, 56.587
Other waste, Bio-conversion, 77.81
Pumped heat, Heating and cooling - homes, 193.026
Pumped heat, Heating and cooling - commercial, 70.672
Solar PV, Electricity grid, 59.901
Solar Thermal, Heating and cooling - homes, 19.263
Solar, Solar Thermal, 19.263
Solar, Solar PV, 59.901
Solid, Agriculture, 0.882
Solid, Thermal generation, 400.12
Solid, Industry, 46.477
Thermal generation, Electricity grid, 525.531
Thermal generation, Losses, 787.129
Thermal generation, District heating, 79.329
Tidal, Electricity grid, 9.452
UK land based bioenergy, Bio-conversion, 182.01
Wave, Electricity grid, 19.013
Wind, Electricity grid, 289.366

```

⌘ + Enter | [Run ▶](#)

Syntax

The idea behind syntax is that a user types **sankey** keyword first, then pastes raw CSV below and get the result.

It implements CSV standard as [described here](#) with subtle **differences**:

- CSV must contain **3 columns only**
- It is **allowed** to have **empty lines** without comma separators for visual purposes

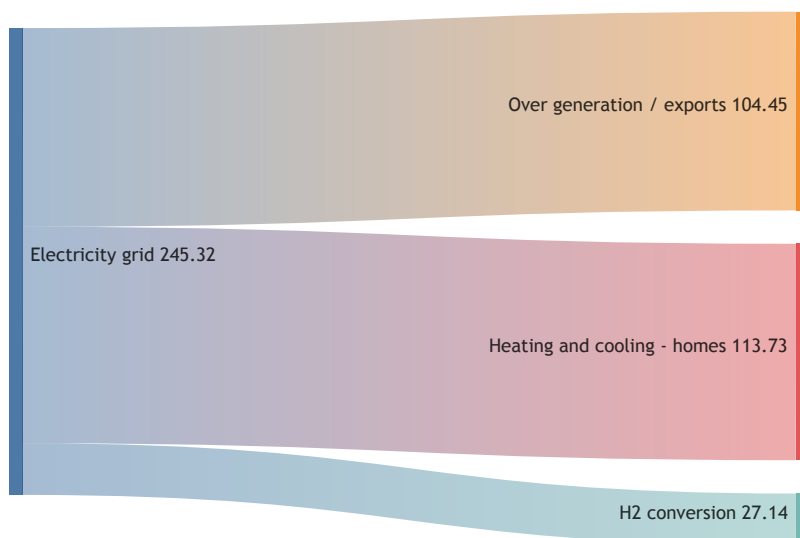
Basic

It is implied that 3 columns inside CSV should represent **source** , **target** and **value** accordingly:

Code:

```
sankey
%% source,target,value
Electricity grid,Over generation / exports,104.453
Electricity grid,Heating and cooling - homes,113.726
Electricity grid,H2 conversion,27.14
```

⌘ + Enter | [Run ▶](#)



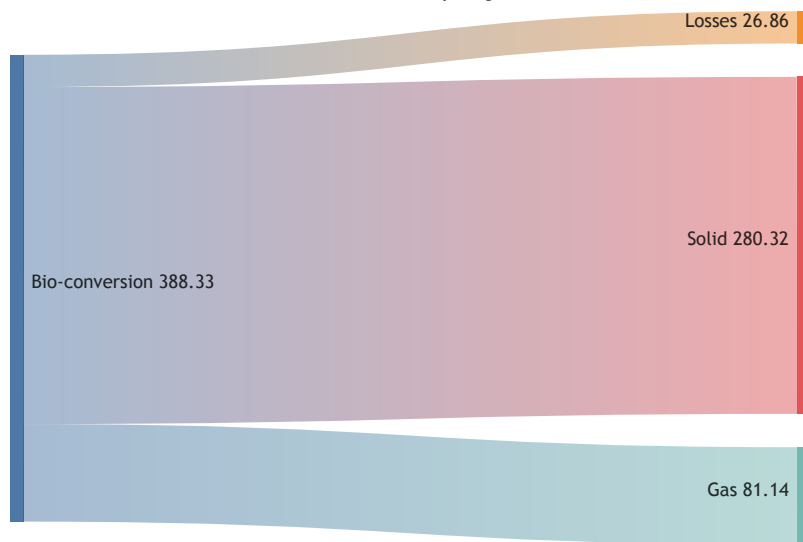
Empty Lines

CSV does not support empty lines without comma delimiters by default. But you can add them if needed:

Code:

```
sankey
Bio-conversion,Losses,26.862
Bio-conversion,Solid,280.322
Bio-conversion,Gas,81.144
```

⌘ + Enter | [Run ▶](#)



Commas

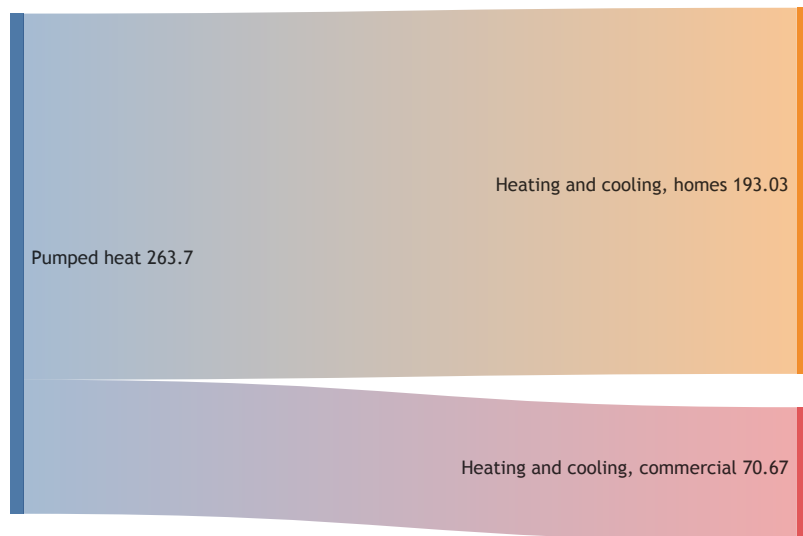
If you need to have a comma, wrap it in double quotes:

Code:

```
sankey
    Pumped heat,"Heating and cooling, homes",193.026
    Pumped heat,"Heating and cooling, commercial",70.672
```

mermaid

⌘ + Enter | [Run ▶](#)



Double Quotes

If you need to have double quote, put a pair of them inside quoted string:

Code:

```
sankey
    Pumped heat,"Heating and cooling, ""homes""",193.026
    Pumped heat,"Heating and cooling, ""commercial""",70.672
```

mermaid

⌘ + Enter | [Run ▶](#)



Configuration

You can customize link colors, node alignments and diagram dimensions.

```
<script>
  const config = {
    startOnLoad: true,
    securityLevel: 'loose',
    sankey: {
      width: 800,
      height: 400,
      linkColor: 'source',
      nodeAlignment: 'left',
    },
  };
  mermaid.initialize(config);
</script>
```

html

Links Coloring

You can adjust links' color by setting `linkColor` to one of those:

- `source` - link will be of a source node color
- `target` - link will be of a target node color
- `gradient` - link color will be smoothly transient between source and target node colors
- hex code of color, like `#a1a1a1`

Node Alignment

Graph layout can be changed by setting `nodeAlignment` to:

- `justify`
- `center`
- `left`
- `right`

[Edit this page on GitHub](#)

Previous page
[ZenUML](#)

Next page
[XY Chart 🔥](#)

